Dirichlet approximation rules out the proposed logarithmic lower bound

Disproof of Conjecture 00000009690

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Abstract

Keep the algebraic arguments fixed at 2 and 3, and use integer coefficients. For every $C > 0$ and every height cutoff H_0 there exist $a > 0$, $b \geq 0$ with actual height $H = \max(3, a, b) \geq H_0$ such that

$$0 < |a \log 2 - b \log 3| < \frac{C}{(\log H)^2}.$$

Thus the conjectured lower bound fails already at $\varepsilon = 1$, even allowing an arbitrary positive implied constant and an arbitrary sufficiently-large-height condition. The nonvanishing, integer coefficient bounds and unbounded actual heights are all proved in Lean 4.

Height convention and the statement being refuted

The standard coefficient-height parameter for integer linear forms is $\max(3, |b_1|, \ldots, |b_n|)$; see [2, abstract]. We use $b_1 = a$, $b_2 = -b$ and fixed logarithms of the positive rational algebraic numbers 2 and 3. Hence their heights and degrees do not vary. Integer coefficients are a subclass of the rational coefficients allowed by the conjecture.

The claimed improvement would in particular provide a positive constant lower bound of order $(\log H)^{-2}$ for these fixed arguments, once H is sufficiently large. We disprove even that version. This concerns the literal logarithmic lower bound in the source, without claiming to determine any correct optimal power $H^{-\kappa}$ in Baker's theory.

Lemma 1. *If $a > 0$ and $b \geq 0$ are integers, then $a \log 2 - b \log 3 \neq 0$.*

Proof. Equality would imply $\log(2^a) = \log(3^b)$ and hence $2^a = 3^b$. The left side is divisible by 2; the right side is not. □

Lemma 2. *For every integer $N \geq 1$ there are integers a, b with $1 \leq a \leq N$, $0 \leq b \leq N + 1$, and*

$$0 < |a \log 2 - b \log 3| \leq \frac{\log 3}{N + 1}.$$

Proof. Put $\xi = \log 2/\log 3 \in (0,1)$. Dirichlet's approximation theorem gives $1 \le a \le N$ and an integer b with $|a\xi - b| \le 1/(N+1)$. One proof considers the $N+1$ points $0, \xi, \dots, N\xi$ modulo 1: one of their circular gaps is at most $1/(N+1)$, and taking the two endpoint indices gives the asserted approximation, including the possible gap crossing 1.

Because $a\xi \ge 0$ and the error is strictly below 1, the integer b is nonnegative. Also $a\xi \le N$ and the error is at most 1, so $b \le N+1$. Multiplication by $\log 3 > 0$ proves the upper bound; Lemma 1 proves strict positivity. □

Theorem 1. *For every $C > 0$ and $H_0 \in \mathbb{N}$ there are integers $a > 0$, $b \ge 0$ such that $H = \max(3, a, b) \ge H_0$ and*

$$0 < |a \log 2 - b \log 3| < C/(\log H)^2.$$

Proof. The nonzero forms with $1 \le a \le H_0$ and $0 \le b \le H_0$ form a finite set. There is therefore $\delta > 0$ below every absolute value in this set; if the set is empty take $\delta = 1$.

Choose $N \ge 2$ large enough that

$$\frac{\log 3}{N+1} < \delta, \qquad \frac{\log 3\,(\log(N+1))^2}{N+1} < C.$$

Both choices are possible by the elementary logarithmic growth limit. Obtain a, b from Lemma 2. Their actual height satisfies $H \le N+1$. If $H < H_0$, both coefficients would be at most H_0, contradicting $|a \log 2 - b \log 3| < \delta$. Thus $H \ge H_0$. Since $H \ge 3$, logarithms are positive and monotone, and

$$|a \log 2 - b \log 3|(\log H)^2 \le \frac{\log 3\,(\log(N+1))^2}{N+1} < C.$$

Divide by $(\log H)^2 > 0$. □

This proves failure for every positive constant and every cutoff at $\varepsilon = 1$. A claim for every $\varepsilon > 0$ is consequently false. The actual maximum height is used throughout; inflating a declared upper bound is not the source of the contradiction.

Formalization and reproducibility

The Lean project uses the actual real logarithm and integer coefficients. The nonvanishing lemma uses injectivity of the logarithm on positive reals and prime divisibility. The approximation lemma invokes Mathlib's proved Dirichlet theorem and establishes coefficient bounds and sign conditions.

`logarithmic_error_limit` and `reciprocal_error_limit` prove the two needed limits. A finite-set induction proves a positive lower bound for the finitely many small-height forms. The final `counterexample` has arbitrary $C > 0$ and H_0 and uses `height a b = max 3 (max a b)`. The theorem `no_uniform_bound` directly negates the existence of a positive constant and a large-height cutoff for the claimed $\varepsilon = 1$ lower bound.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The project is freshly built. The axiom audit contains only standard foundational axioms, with no proof placeholders, custom axioms or external computation oracles.

The optional exact-check script `verify.py` uses only Python's standard library. It constructs rational intervals enclosing logarithms from the convergent series

$$\log u = 2\sum_{j\geq 0} \frac{t^{2j+1}}{2j+1}, \qquad t = \frac{u-1}{u+1}, \quad 1 \leq u \leq 2,$$

with a geometric remainder bound. It uses rational continued fractions to find examples, then checks their nonzero signs and error inequalities with exact fractions. Floating point is used only to display the already bounded errors. Finite examples do not prove the universal statement or a limit; those are proved by Lean. Run `python verify.py` and `tectonic main.tex`.

The exact bilingual conjecture is in `SOURCE.md`; provenance, real build logs, file hashes and PDF review are in `verification/`. The author performed an adversarial self-review. No independent reviewer or subagent was used. The argument is an application of classical approximation, not a claimed new result about linear forms in logarithms.

References

[1] The Last Math Competition, Conjecture 00000009690. Exact source and upstream provenance are included with this submission.

[2] Y. Bugeaud, B', 2022, arXiv:2209.00275, abstract (coefficient-height convention and background).

[3] The Mathlib community, *Diophantine Approximation*, Dirichlet approximation theorem at the commit pinned above. Lean source.